
Software Engineering

— CSE-416-03 · Fall 2026 —

Staff

- Lecturer: Dr. Shuai Mu
 - Please address me as Shuai (start your email as “Hi Shuai, ...”)
 - shuai@cs.stonybrook.edu (the @stonybrook.edu address also works, with a delay)
- TA: to be announced

Office hours

- MW 1–2pm, NCS 351 — right before class
- Book a slot on [GCal](#) (at least a day ahead if you want a locked-in time)
- Class is MW 2:00–3:20pm in NCS 120
- After class I go to another course on West Campus, so “walk with me after class”
will not work this semester

Piazza (for announcements, not debugging)

- <https://piazza.com/stonybrook/fall2026/cse41603>
- Join today. Course announcements go there.
 - If I post a broadcast, the assumption is you will read it within 24 hours.
- Do **not** treat Piazza or Stack Overflow as your first stop for coding help
 - That workflow is from a previous era
- Stuck on JavaScript, CSS, a stack error, a weird build? Ask an AI.

Class website

- <http://mpaxos.com/teaching/se/26fa/>
- Syllabus (also as PDF), schedule, office hours
- Updated as the semester goes

A little history of the class...

Era 1: J2EE + UML

- The old 416: Java, J2EE, and a lot of UML
 - Use-case diagrams, class diagrams, sequence diagrams — then code
- Lectures taught **the** process and **the** stack
- You designed on paper (or in a modeling tool) before you shipped
- A fine way to learn vocabulary. A slow way to ship a product.

Era 2: Python / Node / MERN

- Then the class became a web-app class
 - Django, Flask, Node, React, Mongo — MERN and friends
- Still a prescribed stack. Still a lot of lectures on how to type it.
- You built **a** web app, not **your** product
- Faster than J2EE. Still last decade's default internship stack.

This year is different

- Those two eras assumed you would type the stack by hand
- Last year: one shared project (an IPFS-like blockchain) in a fixed stack
- This year: **you** pick the product
 - Web or desktop; no required domain, architecture, or tech stack
 - Build it the 2026 way: agents, not hand-typed homework

What will we do in this class?

- Prepare you for a software-engineer job **in 2026**
 - Modern Internet companies (FAANG-style and startups)
 - Two overall abilities: **hack** (with AI) and **communicate** (teamwork)
 - Ship a product, not a pile of hand-typed homework
- We will do this by **PRACTICE**
 - A semester-long project with a team, of your own choosing
- Lifecycle of a project
 - Find what you want to do → design → implement → test → ship
 - Learn, discuss, and review at each step

What will we NOT do in this class?

- We will not teach you to type JavaScript / CSS / Python by hand
 - We are not going to pretend we can beat AI at that
- We will not debug your specific setup problems
 - e.g. “my Python 2 library does not work with Python 3 — can you help?”
 - Ask an agent. Paste the error. Iterate.
- There are no exams, quizzes, or a required textbook

Requirements

- Officially: C or higher in CSE 316; U4 standing; CSE major
- Informally: you should be almost ready to work as an engineer
- You can steer a coding agent
 - You do not need 12 weeks to “learn a language” before you can ship in it
- CSE 305 / 333 / 336 / 337 / 356 are **not** required
 - Need a database, a UI, a new stack? Have the model help you pick it up.
- If your instinct is still “I should write this CSS myself” — unlearn that

The project is the course

Class project

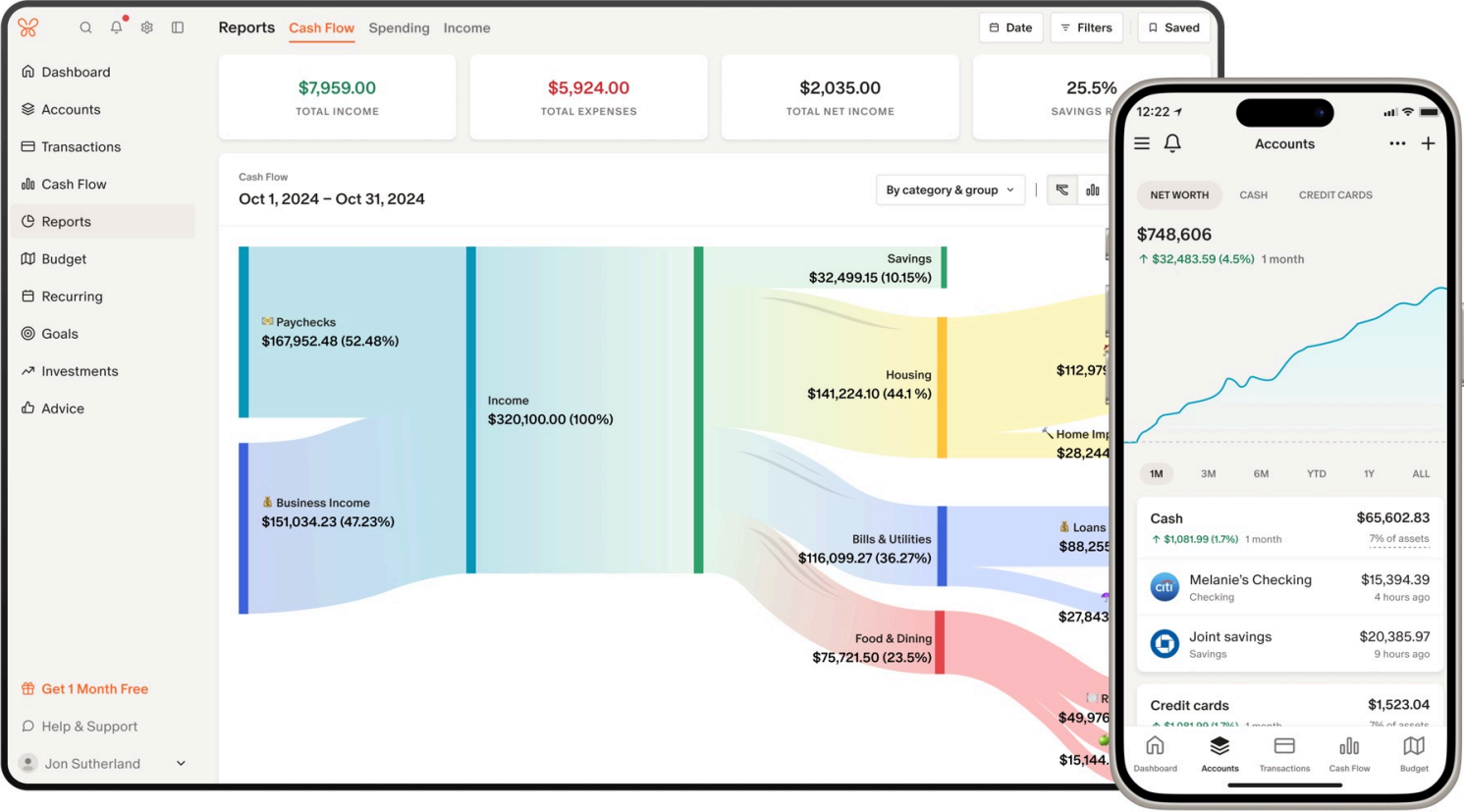
- Pick something you actually want to ship by December
 - Web, desktop, or mobile. We will **not** assign a topic, stack, or architecture.
- It has to be a **product** with real users — not a one-purpose tool
 - Accounts; login; something they actually do after they are in
 - Several screens, shared data, more than one kind of user or workflow
 - AI makes coding faster; it does not turn a thin page into a product
- You should be able to say who the users are and why they come back
- Do not want to invent a product? Two staff projects below. If unsure, ask **before** you lock it in.

You do not have to invent the idea

- Eric Schmidt (ex-CEO of Google) to Stanford students, 2024:
 - Tell your LLM: “**Make me a copy of TikTok ... produce this program ... release it.**”
- That is a fine way to pick **this** project
 - Choose a popular product people already use. Clone the **idea**.
- Build **your** version: your code, your name, your users
 - Do **not** take their logo, music, trademarks, or data

Example: Monarch

Money app — accounts, transactions, reports.



Example: AwardWallet

Loyalty / trips — many accounts, not a scraper page.

The screenshot displays the AwardWallet web application. The top navigation bar includes links for Accounts (8), Trips (10), Bookings (1), Blog, Community, FAQs, APIs, Promos, Credit Card Offers, and Contact Us. The user profile for John Smith is visible on the left sidebar, along with options to add award programs and track manually. The main content area is divided into three sections: Airlines, Hotels, and Shopping. Each section lists various loyalty accounts with details such as the program name, account number, status, balance, and expiration date. A summary table at the bottom of each section shows the total number of accounts and the combined balance. The Airlines section lists accounts for Alaska Airlines, Lufthansa, British Airways, JetBlue, and Jet Blue. The Hotels section lists Hilton, IHG, and Marriott. The Shopping section lists My Best Buy. A footer at the bottom contains copyright information, privacy policies, and mobile app download links for Android and iPhone.

AwardWallet Accounts 8 Trips 10 Bookings 1 Blog Community FAQs APIs Promos Credit Card Offers Contact Us

John Smith English Add Award Program to Track

PERSONS 8

ALL 8

John Smith 7

Jennifer Smith 1

Manage Connections 2

MANUALLY TRACK

Award Program

Gift Card

Travel Voucher

INVITE TO AWARDWALLET

Invite using a direct link: <https://awardwallet.com/ref/C>

Or:

Your friend's email here...

Free upgrade coupon for the first time users: free-coupon 10 left

Like 5.5k Share

Invite Twitter followers

People Accepted: 5

Available Upgrades: 1

Referral Bonus Points: 0

AwardWallet OneCard

Get your personal AwardWallet OneCard with all of your account numbers on it

Airlines

Award Program	Account	Status	Balance	Expire
JOHN SMITH				
Alaska Airlines (Mileage Plan)	1234567	MVP Gold	102,412 +1,000	in 1.9 years
Special Fare in 5.3 months				
Lufthansa (Miles and More)	JSmith@gmail.com		9,000	
British Airways (Executive Club)	543210	Silver	46,423	in 2.9 years
JetBlue Airways (trueBlue)	JSmith@gmail.com	Member	52,300	
Jet Blue (Travel Bank)			\$1,200	

Hotels

Award Program	Account	Status	Balance	Expire
JOHN SMITH				
Hilton (Honors)	JSmith	Gold	37,697 +2,500	in 10.6 months
IHG One Rewards	JSmith@yahoo.com	Silver Elite	33,326	in 2.6 months
Food & Beverage Reward in 4 months				
Marriott Bonvoy	JSmith@yahoo.com	Titanium Elite	7,268	in 1.8 years

Shopping

Award Program	Account	Balance	Expire
JENNIFER SMITH			
My Best Buy	JenSmith@gmail.com	224	in 2.5 months

Totals: 8 accounts 280,850

Add Award Program to Track

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Android iPhone

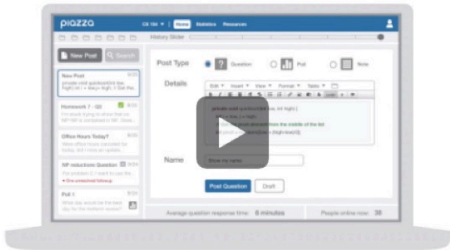
Example: Piazza

Q&A / discussion — accounts, posts, a real class.

 [Product](#) [In Professors' Words](#) [Pricing](#) [Support](#) [About Us](#) [Piazza Talent](#) [Sign Up](#) [Login](#)

The incredibly easy, incredibly engaging Q&A platform

Save time and help students learn using the power of community



- Wiki style format enables collaboration in a single space
- Features LaTeX editor, highlighted syntax and code blocking
- Questions and posts needing immediate action are highlighted
- Instructors endorse answers to keep the class on track
- Anonymous posting encourages every student to participate
- Highly customizable online polls
- Integrates with every major LMS

[Students Get Started](#) [Professors and TAs Get Started](#) [View a Real Class](#)

[Learn more about how Piazza complies with FERPA](#)

New to Piazza? Watch these videos to get started:

[Creating and configuring Your Class](#)
[Announcing Piazza to students](#)

[Posting your first note](#)
[Piazza intro for students](#)

Our team is on standby to help you get started with Piazza. Email us at team@piazza.com if you need help!

Over 100,000 professors have chosen Piazza. Millions of students in thousands of universities in 90 countries are using Piazza.

Example: Stardew Valley

A video game.



What does not count

- A thin utility page, even if it technically has visitors
- A campus map of Stony Brook, a torrent search site, or any similar one-purpose tool
- **Not allowed** (morality): gambling, adult content, prediction markets (e.g. Polymarket-style betting)
- If you are unsure, ask

Backup: work on a staff project

- Some of you would rather **not** invent a product from scratch
- You can join a team that works with me on a systems project instead
- Two options. Both are **huge**. That is the point, if you like systems.
- Same milestones, same grading. You still have to ship and explain.

Chiral Network — A blockchain

- Last year's project, continued
- A **blockchain for resource sharing** — not just a token
 - Exchange resources (storage, compute, bandwidth, ...)
 - Run hosted functions on the chain (think Internet Computer canisters)
- If you have ever wanted to know how that kind of network is actually built

Option B: a cloud database

- A cloud database in the spirit of **Supabase**
 - Auth, storage, a query API, the boring infrastructure everyone uses
- Postgres-shaped, cloud-hosted, meant to be a real backend for apps
- If you care about how a production database service is put together

Staff projects: the deal

- You work with me; we will not pretend this is a weekend app
- You will learn a lot **if** you are interested in systems
- Say so when you form teams (Aug 26 / 31). Limited seats.

a student life vs. an engineer one?

a course vs. a real project?

This class is closer to the right-hand side.

You will not be told exactly what to build. You **will** be asked to scope, design, ship, and explain the choices.

This class will be very different from your past classes...

- No textbook, no quizzes, no exam
- No one will tell you exactly what you must (or must not) do
- You will need to work with others
- You will need to **read and explain** code — including code a model wrote
- Use AI as much as you want. That is the default, not a loophole.
- Grades come from what the team **ships**, how you contributed, and one group presentation — not from a curve of exam scores

Find your teammate

- Form a team during the Aug 26 and Aug 31 sessions
 - It is **your** responsibility to join a team
- Aim for about 5–6 people
 - Ten presentation topics; one talk per team
- Can I work alone? Yes and no
 - This is a **team-project** class, not just a project class

A few suggestions on finding teammates

- Communication
- What type of person are you?
 - A hacker? A leader? An “HR”?
- Non-technical reasons matter too
 - Where do you work (on-campus? off-campus?)
 - When do you work (morning person or evening person?)
 - Why do you work (just pass, or maybe turn this into a startup?)

How will we spend time in class?

- A few lectures
 - Concepts that matter; nothing to memorize
 - We will not go into the specifics — read the docs when you use a tool
- Student presentations (one per team)
 - A **story**, not a textbook recap
- Milestone reviews
 - About 10 minutes: what you finished, why, how the team (including AI) worked
- Week 1: two free sessions to team up and brainstorm

The shape of the semester

Lectures

Aug 24 Overview (today)

Sep 2 SE tech overview

Sep 9 Working with agents

Sep 7, Oct 12, Nov 25 no class

Presentations (10 topics)

Sep 21 → Nov 23

Milestones (in class)

Sep 14–16 M1 Proposal

Sep 28–30 M2 Design & setup

Oct 19–21 M3 MVP

Nov 2–4 M4 Feature-complete

Nov 16–18 M5 Testing & integration

Nov 30–Dec 2 M6 Final demo

Dec 7 Poster session

Class presentations

- One **30-minute** talk per team; everyone speaks
- Split one big story, or each person tells a smaller story on the same topic
- Tell a **story**: how a company, an incident, or a real project actually does this
 - What they do, what it is for, what happened
 - Interesting failures count
- Strongly discouraged (you would get a low grade):
 - “What is X”, taxonomies, concept dumps
 - Assume the class can look those up
- Send slides at least 72 hours before your slot
- Topics assigned after teams are set

How am I evaluated?

- **Course project — 80%**, in six milestones
 - M1–M5: 10% each
 - M6 Final product: 30%
- **Class presentation — 20%**
- No exams
- Milestone grades look at the quality of what the team delivers **and** each student's contribution
- Teams are graded on the day they present
- We will ask you to assess your teammates at milestones

What a milestone review looks like

- About 10 minutes
- Cover (1) what you finished, (2) the design choices and why, (3) how the team — including AI — worked in this phase
- Have the artifacts in the repo (proposal, design, tests, ...)
- Bring a short demo or walkthrough

Override: ship something people actually use

- One way around the usual milestone grading
- If the product has **real** daily active users for **at least two weeks**:
 - 1000+ DAU → course grade **A**
 - 500+ DAU → course grade **A–**
- We will check the numbers
 - Classmates clicking a link, bots, and fake accounts do not count

We live in a revolutionary age of AI

Use AI as much as possible

- This is not “allowed.” This is how the work gets done.
- Autocomplete (Copilot) is the floor, not the ceiling
- Agentic tools (Cursor, Claude Code, Codex, ...) are better
- Vibe-code if you want. Loop-code if you want. Spec → generate → run → paste the error back → repeat.
- Do **whatever you think is productive**

Do not write the code by hand

- Do not sit there typing JavaScript, CSS, HTML, or glue code like it is 2018
- We are not going to pretend a student (or the instructor) beats a frontier model at that
- Your job is to **direct** the work: what to build, whether it is right, what to try next
- If you are grinding out a stylesheet by hand, you are practicing the wrong skill

Piazza / Stack Overflow are the old world

- Searching for an error, posting a snippet, waiting for a stranger — that loop is slow now
- First: paste the error into an agent and let it iterate
- Piazza is for **course** news, not for “how do I center a div”
- Staff will not hand-debug your toolchain

What AI does not replace

- You still have to **understand** what shipped
 - If you cannot explain it to a teammate, you did not do the work
- You still decide **what** to build and **why**
 - The model implements; it does not pick the product
- You are still on the hook for correctness, security, and quality
- Disclose AI use the way you would cite any other source

TODO this week

- Join [Piazza](#) today
- Wednesday (Aug 26): a **3-minute** lightning talk on the ideas you want to do
- Aug 26 and Aug 31: form a team and brainstorm a project
- Lock in something you actually want to build
- Once the team exists, share a GitHub (or similar) repo with the staff
 - Details in class
- Presentation topics will be assigned after teams are set

Attendance

- No one will take attendance on ordinary lecture days
- Milestone days and your presentation day **are** the grade — be there, ready to demo
- Use AI as much as you want, on everything
- Look at any online materials that are legal (do not hack into them; follow the license)

Penalty

- You will receive an F if you
 - Copy another team's project, presentation, or writeup
 - Claim someone else's work as yours
 - Sabotage this class or others' work
- Please do not negotiate for more scores or extra work
 - Each attempt is a 10% penalty on the overall grade, and it stacks
- I hope we do not need to apply this. I will.

General advice: hardware

- Dev machine: any modern laptop is fine
- You will need to **demo** a running system at milestones
 - Local is fine early; later milestones want something teammates (and we) can actually run or visit
- A cloud VM is useful if you ship a web service, not required on day one

General advice: choosing toolchains

- Pick a stack your **agents** are good at, and that the team can demo
- Language-first or framework-first — either is fine
 - Python then Django, or Rails then Ruby, or whatever ships
- Do not choose a stack because you want to “learn it the hard way” in this class

General advice: choosing toolchains (cont'd)

- Better to avoid
 - Toolchains that lock you to one platform (e.g. C#)
 - Tools with pretty demos but no real community
 - Tools that were once popular and are now unmaintained
- “Is the model good at this stack?” is a fair question now
- There is no required stack. Pick something the **team** can ship with AI, not something you plan to type by hand.